



Pulkovo observations of apparent close approaches between near-earth asteroids and the Gaia stars in 2019–2020[☆]

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ABSTRACT

The astrometric observations of the apparent close approaches between near-Earth asteroids and the Gaia stars provide high quality data for studies on the dynamics of these objects. We present the **technique of data analysis that allows us to decrease the distortion-dependent positional errors** and observational results of eight of such events performed with the 'Saturn-1m' telescope of the Pulkovo Observatory in 2019–2020. The mean value of the astrometric accuracy of the relative 'star - asteroid' positions is ± 30 mas.

1. Introduction

Statistical analysis of the orbital parameters of the near-Earth asteroids (NEAs) and the research of their evolution allow scientists to make progress in the understanding of the Solar System dynamics and development. The numerical experiments make it possible to estimate typical lifetime of the NEAs (about several million years) (Farinella et al., 1994). Hence, the analysis of the NEAs dynamics will help to determine presumable mechanisms of the NEAs population replenishment. It is well known that the chaotic-like motion, resonant locking, and the Yarkovsky–O'Keefe–Radzievskii–Paddack effect (YORP) effect have the largest impact on the orbital evolution of the NEAs (see review paper published by Morbidelli et al. (2002)). Therefore, accurate measurements of the NEA positions will show the distribution of these effects over the mentioned physical mechanisms.

Nowadays the most accurate positions and velocities of the NEAs are obtained with the radar or the space-based observations. Unfortunately, these data are available for a small part of the NEA population. As a result, ground-based astrometric observations deliver a significant part of the required data. Traditionally the equatorial coordinates of an asteroid in the field of a CCD-image are determined using calibration with the reference stars from an astrometric catalog. The majority of the NEAs are relatively faint ($\text{mag} > 16.0$) and fast (apparent motion rate is about several arcmins per hour). **The unaccounted optical distortion and various atmospheric effects (such as refraction and atmospheric dispersion) provide additional difficulties. This leads to a positional accuracy of about 100 mas – 200 mas or worse using the pre-Gaia astrometric catalogs (Vereš et al., 2017).**

Significant refinement of the astrometric accuracy of the NEAs observational data became possible thanks to the Gaia mission. The second release of the Gaia data (Gaia DR2) contains about 300 thousand observations at the several mas level of accuracy (in along scan direction) for 14099 **Solar System** objects (Gaia Collaboration et al., 2018). The application of the Gaia data for the analysis of the NEAs dynamics is considered in Bancelin et al. (2012) paper. However, the Gaia data cover a relatively small time interval (22 months for Gaia DR2) and include a small part of the NEAs population. **The newest release of the Gaia mission (Gaia EDR3) doesn't contain the results of asteroid observations (Gaia Collaboration et al., 2020).** Hence, the ground-based astrometric observations of the NEAs in the Gaia reference frame are necessary.

Gaia DR2 and Gaia EDR3 provide sub-milliarcsecond precision for the stars traditionally used as reference objects for astrometric reductions of the CCD-frames containing the NEAs. Hence, the uncertainty of the NEAs coordinates extracted from such observations depends on the accuracy of the asteroid image fitting if all systematic effects are taken into account. We expect the NEAs positional uncertainty to reach the level of 20 mas – 50 mas or lower. However, the analysis performed by Vereš and colleagues shows that this isn't true for a wide set of telescopes usually used for astrometric observations of the NEAs (Vereš et al., 2017). Typical accuracy is about 100 mas as stressed in Zhai and coauthors paper (Zhai et al., 2018).

This motivates astronomers to look for methods to improve the accuracy, such as synthetic tracking technique based on series of short-exposure images (Zhai et al., 2018). Another possible way to make

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improvements in accuracy is to conduct observations of the stellar occultations caused by NEAs. This gives the NEA positions at the 1 mas level of accuracy, but such events are relatively rare for a certain observing site and a certain object. In this regard, an idea to observe apparent close approaches between the Gaia stars and the Solar System bodies is an adequate alternative for when a systematic distortion of the telescope projection is the main reason for the positional bias of the NEAs. It is important to note that the projection-dependent systematic displacement is nearly the same for both the asteroid and the star images during the apparent close approach between them. The success of the implementation of this technique depends on the frequency of the events and their distribution over the celestial sphere. The reference stars of the Gaia catalog have unprecedentedly high positional accuracy and a uniform density. These two facts guarantee perfect conditions for capturing the required number of events (Bikulova, 2019).

The first stage of the observational program of the apparent close approaches of selected NEAs to the Gaia stars was performed at the Pulkovo observatory. Our goal was to test the technique with observations of several events and look for ways to improve a future realization of the wider program. This paper summarizes our experience. The technique described in our paper isn't new. The section 2 contains other examples of its implementation and references. The motivation to use the apparent 'NEA – star' close approaches is discussed in Section 3. The short description of the observations and the data analysis is presented in Section 4. Section 5 contains the results and their comparison with the ephemerides. The final conclusions are given in Section 6.

2. Some remarks on the application of the apparent close approach technique

This technique is used frequently in astrometry. A common reason for the application of the method is the absence of the reference stars in the small field of view of the telescope. For example, high-angular resolution observations of close approaches between stars and asteroids using adaptive optics systems are mentioned in Perryman's book (Perryman (2009), pages 573–575). The paper Morgado et al. (2019) gives a good example of how to use the events of close approaches between two Jovian satellites to improve the astrometric accuracy of the positions of the Galilean moons without the reference stars. The apparent close approaches of the NEAs to the reference objects allow us to link the dynamically defined reference frame (ephemerides) with the kinematic realization of the reference system by the astrometric source catalog. The analysis of the mutual orientation of these reference frames can be performed using the results of the observations of the considered events. This idea was brought in by Nedelcu et al. (2010) for the apparent close approaches of the NEAs to ICRF quasars. As mentioned in the Introduction, our interest in the close approaches between the Gaia stars and NEAs is caused by another reason. Here we attempted to show that the technique of observations of apparent close approaches between stars and asteroids is not new for ground-based astrometry.

3. The motivation to use the apparent 'NEA – star' close approaches

The Gaia EDR3 catalog provides the data for tens (or more) of stars in the 10-arcmin field of view (FOV). The positions of most of the Gaia stars are known with an excessive precision (better than 1 mas) with respect to the positional standard errors of the CCD-measurements. The accuracy of the extraction of pixel coordinates x , y from the stellar images lies within 0.001–0.1 of a pixel size depending on the signal-to-noise ratio. It means that it is easy to form a set of the reference stars with formal standard errors of pixel coordinates about 10–50 mas in the 10-arcmin FOV. Therefore, it seems obvious that the initial accuracy of stellar image

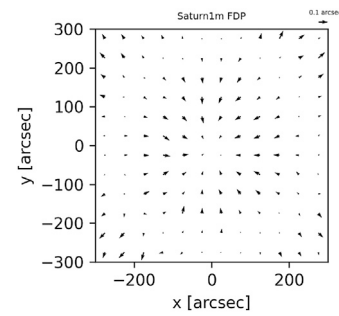


Fig. 1. Field distortion pattern¹ for the Pulkovo 'Saturn-1m' telescope. Each vector is a result of averaging the tangential coordinate residuals calculated using a linear model of transformation from pixel to tangential coordinates. The averaging is performed for the hundreds of the residuals of the reference stars located in the corresponding small box in the FOV and accumulated from analysis of thousands of the CCD-frames.

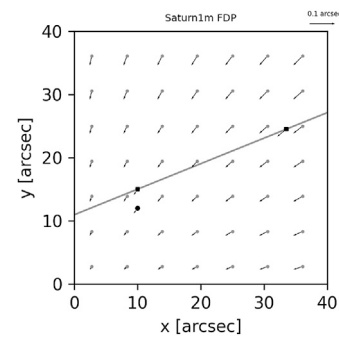


Fig. 2. A sketch that illustrates the basic idea of application of the astrometric observations of apulsers considered to improve the positional accuracy.

measurements could be maintained only if there was a simple linear model of transformation from pixel positions to the tangential coordinates. The expected accuracy could be achieved in the absence of systematic projection errors, but they always exist for telescopes used for mass observations.

A typical map of systematic errors (field distortion pattern, FDP) is shown in Fig. 1. As can be seen in this figure, positional residuals reach 100 mas. A technique of positional correction calculation based on FDP construction is well known. The paper of Zacharias (1995) provided a good example of the method implementation.

In a relatively dense stellar field provided by hundreds of the reference stars a pixel-to-tangential coordinate transformation can be represented with the terms of the second-, third- or higher-order. This allows us to avoid the FDP construction. However, mostly there are 10–20 Gaia stars in the typical FOV. This number of stars do not provide suitable coverage of FOV to use high-order terms in astrometric calibrations. Therefore, it is hardly surprising that the accuracy of the 'classic' NEAs astrometric observations is about 100 mas in the Gaia epoch.

The natural way to solve the problem of unaccounted systematic effects is the calculation of corresponding corrections. The FDP can be formed as a result of the analysis of thousands of CCD-frames and the necessary correction can be calculated. This solves the problem for stable FDP. In real cases, FDP depends on observational conditions (air temperature, telescope position, stellar magnitudes, etc.). This makes it difficult to use the FDP-based corrections.

Another way to avoid the already mentioned systematic errors is based on the smoothness of the FDP. In this case, the FDP-based correction should be the same for all objects in a small area in comparison with the FOV size. This is realized for close apparent celestial objects like binary star components, planetary satellites, and of course for

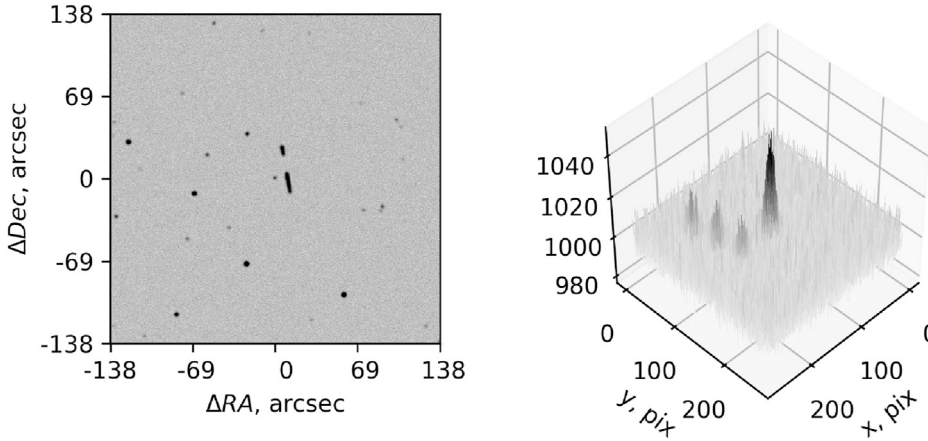


Fig. 3. An illustration of a typical event. The top panel contains a stacked image of 60 CCD-frames centered at the star with Gaia source id = 3436496968115120512, $G_{\text{mag}} = 16.223$. Two tracks near the center were generated by separate images of the NEA 333555 (2005 VY17) ($\text{mag} = 16.4$) projected to the single CCD-frame. The exposure time of each CCD-frame is 5 s. The break between the tracks is due to weather conditions. The bottom panel shows the detailed structure of the image from the central part of the top panel. The central peak is the star image. Three separate peaks are examples of asteroid images projected from separate CCD-frames.

asteroids during their close approaches to stars. As a result, it might be possible to obtain ‘NEA – star’ relative positions with the accuracy of pixel measurements.

The illustration of the advantages of the astrometric observations of the events considered in contrast to traditional technique is presented in Fig. 2. A fragment of the real FDP from Fig. 1 is shown here. A black dot corresponds to the virtual Gaia star, two black squares simulate two positions of the NEA. Corresponding vectors of distortion-dependent systematic biases for the star and NEA positions were calculated by FDP interpolation. A straight gray line corresponds to the asteroid motion. It is easy to see that the systematic offsets for star and NEA are almost equal to each other at the point of the maximum apparent close approach and their difference is significant at the point that is relatively distant from the star.

Therefore, it is natural that we calculate not only the NEA position using the reference stars and a linear reduction model. We perform the same for the Gaia star that is considered as a target of appulse. Then we calculate the ‘NEA – star’ coordinate differences. If our assumption is correct, we expect to obtain the ‘NEA – star’ position that is almost free of the projection-dependent systematic offsets. The analysis described above shows that only points of the NEA trajectory located near the point of the closest approach to the star can be taken into account to exclude the systematic bias considered.

Of course, our assumptions are limited by a magnitude-dependent positional systematic error. In this connection, it is appropriate to observe only appulses for which the value of the ‘NEA – star’ magnitude differences is less than 1–2.5 mag.

4. Observations and data processing

4.1. Details of the observations

The ephemerides of the apparent close approaches between near-Earth asteroids and the Gaia stars are regularly published on the Pulkovo NEOs web-page¹ using EPOS software (L’vov and Tsekmeister, 2012). This ephemerides contain the events for which the magnitude differences between a star and an asteroid is relatively small (1–2.5 mag). Observations of eight of these appulses were realized with the ‘Saturn-1m’ telescope ($D = 1 \text{ m}$, $F = 4 \text{ m}$) of the Pulkovo Observatory. The ephemerides of the observed events are shown in Table 2.

The ZWO ASI1600 Pro camera was installed at the focal plane of the ‘Saturn-1m’ telescope. The field of view is $13 \times 10 \text{ arcmin}$ and the scale is 172 mas/pix . To robust the determination of the event parameters, the imaging was performed during tens of minutes symmetrically to the central moment of the event (t_c , the moment of time when the angular

distance (ρ) between a star and an asteroid is minimal (ρ_{min})).

The NEAs usually have relatively high apparent motion rates. Hence, their images are seen as tracks in the case of long exposure time. Thus, at this stage of observations the exposure time was reduced to 5 s to get stellar-like images of the NEAs with acceptable values of signal-to-noise ratio.

Fig. 3 shows features of a typical apparent close approach. The right panel of this figure shows the quality of the images of the event star and the asteroid ($\text{FWHM} \approx 3 \text{ arcsec}$).

4.2. Calculation of the coordinates of stars and the asteroid

The shapelet decomposition of the stellar images was used to construct the stellar PSF (Refregier, 2003; Khovrichiev et al., 2018). The shapelet coefficients of the PSF are calculated as median values of the corresponding coefficients of the decomposition of the separate star images. Images of only ten stars located close to the event star were used.

The background value, the flux, and the pixel coordinates x, y were adapted to minimize the norm of the ‘real stellar image - scaled and positioned PSF’ residuals using Nelder-Mead algorithm (Nelder and Mead, 1965). Fig. 4 shows a typical raw stellar image and the result of the shapelet-based analysis. Analysis of the set of the images of the same stellar pairs located in a series of the CCD-frame shows that the pixel coordinate accuracy is about 10–20 mas (the corresponding accuracy estimates are presented in the last two columns of Table 1). More detailed description of the technique test is available in Subsection 4.3.

The pixel coordinate differences between the event star and the asteroid image $\Delta x, \Delta y$ were calculated and converted to $\xi = \Delta \alpha \cos \delta, \eta = \Delta \delta$. Astrometric calibration of the CCD-frames were performed using linear model. The Gaia EDR3 was used as a reference catalog.

These relative coordinates ξ, η can be converted to equatorial coordinates $\alpha_{\text{event}}, \delta_{\text{event}}$ using $\alpha_{\text{star}}, \delta_{\text{star}}$ of the event star from the Gaia catalog. Fig. 5 shows an example of comparison between the (O–C) in Dec values calculated for $\alpha_{\text{event}}, \delta_{\text{event}}$ and (O–C) that represents the results of the standard astrometric reduction process with the reference star from the Gaia catalog.

4.3. Test of the technique using pairs of the gaia stars

As mentioned in Subsection 4.2, a test of the appulses technique was performed. We formed a set of stellar pairs whose images were measured using the same CCD-frames as those analyzed to obtain the appulses results. We excluded selected stars from the list of the reference stars and considered them as ‘asteroids’. Hence, it was easy to calculate celestial position of these stars, corresponding ‘(O–C)’ values using the Gaia EDR3 coordinates of both stars ($\Delta \xi_1$,

¹ <http://www.gaoran.ru/personal/neo/eng/approach/main.htm>.

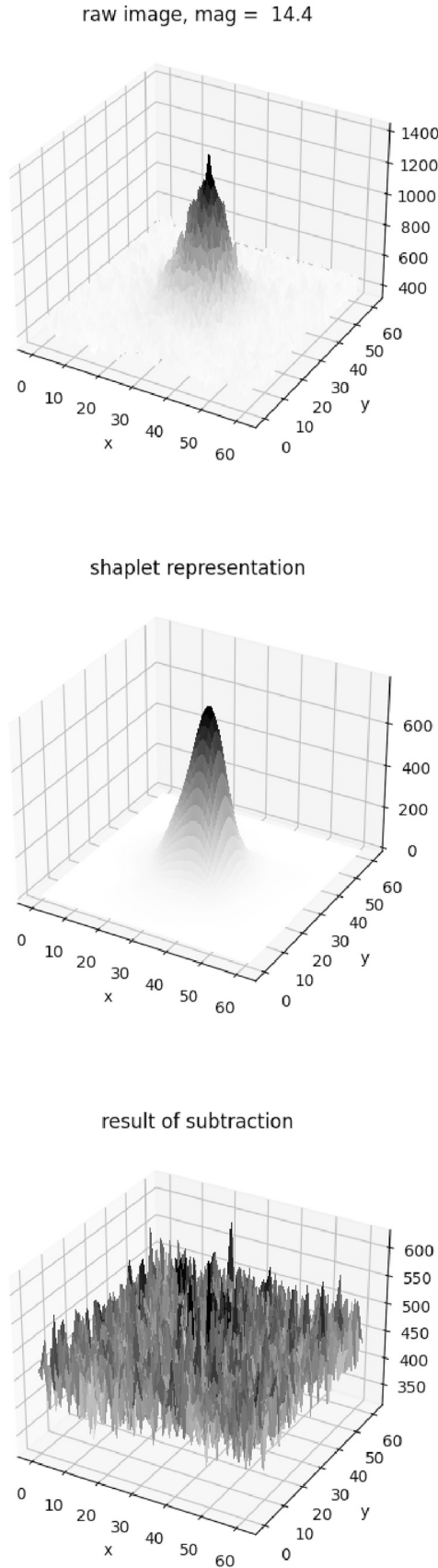


Fig. 4. Demonstration of the quality of the shaplet decomposition of the typical stellar image. The top panel shows the raw image, the middle panel contains the image constructed using the shapelet coefficients, and the bottom panel is the result of subtraction of the reconstructed image from the raw one.

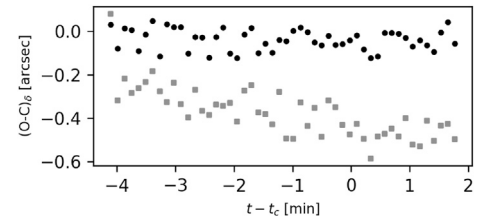


Fig. 5. The $(O-C)_\delta$ as a function of time relative to the moment $t_c = 2019-11-24T19:24:44.9$ for 162723 (2000 VM2). Black dots correspond to the measurements performed relative to the star (Gaia source id = 1942346874552792320, $G = 15.422$) using the apparent close approach technique. The $(O-C)$ values obtained using the 'traditional' astrometric reduction with the reference stars are shown as gray squares. JPL's HORIZONS system (<https://ssd.jpl.nasa.gov/horizons.cgi>) and an astroquery-based script were used to calculate ephemerides.

$\Delta\eta_1$ for the first star and $\Delta\xi_2, \Delta\eta_2$ for the second star). The relative positions ξ, η were considered as a model of relative 'NEA – star' coordinates. Comparison with the Gaia EDR3 data allows us to calculate ' $(O-C)$ ' values for ξ, η ($\Delta\xi, \Delta\eta$) and their standard errors (σ_ξ, σ_η). Table 1 contains the results of calculation.

As can be seen from Table 1, the absolute values of $\Delta\xi_1, \Delta\eta_1, \Delta\xi_2, \Delta\eta_2$ are systematically bigger than $\Delta\xi, \Delta\eta$. In other words, the relative coordinates are in better agreement with the Gaia data than coordinates calculated using the traditional technique. The $\Delta\xi_1, \Delta\xi_2$ values in the last three rows of Table 1 are significant with respect to the values of their standard errors. This allows us to find a match between the basic idea (see Fig. 2) and the results of the analysis of the real CCD-frames. We see that the difference between $\Delta\xi_1, \Delta\xi_2$ values lies within the standard errors range. Both stars of each pair are similarly shifted by the distortion. That is exactly what we expected for a close location of the star and the NEA image during their apparent close approaches. As a result, we conclude that the discussed technique has clear advantages compared to traditional astrometric reduction of the CCD-frames.

4.4. Determination of the event parameters

The motion of an asteroid near the central moment of the event (t_c) is considered as a linear motion with time

$$\xi = \xi_0 + \mu_\xi(t - t_0), \eta = \eta_0 + \mu_\eta(t - t_0).$$

Here ξ_0, η_0 are the relative coordinates of the asteroid at the initial moment t_0 (this is the central moment of the first CCD-frame of the series), μ_ξ, μ_η are apparent **velocities** of the asteroid in the corresponding directions. The values of $\xi_0, \eta_0, \mu_\xi, \mu_\eta$ are calculated by the LSM. These parameters allow us to find the t_c , relative coordinates at this moment ($\xi(t_c), \eta(t_c)$), and their standard errors.

5. The results and comparison with the ephemerides

The $\xi(t_c), \eta(t_c), t_c$ values represent the information about the asteroid motion extracted from the apparent close approach observations. The values of these parameters for each of eight events are presented in Table 3. On the other hand, these parameters of the event can be calculated with the ephemerides (see Table 2). Formally they can be expressed as:

$$\Xi(T_c) = (\alpha_{NEA}(T_c) - \alpha_{star})\cos(\delta_{star}), \\ H(T_c) = \delta_{NEA}(T_c) - \delta_{star}.$$

Here $\alpha_{NEA}, \delta_{NEA}$ are equatorial coordinates of the NEA calculated using ephemerides, $\alpha_{star}, \delta_{star}$ represent the Gaia position of the event star, T_c is the central moment of the event (when 'NEA – star' angular separation reach to minimum value).

Table 1

The results of the test of the appulse technique performed using Gaia stellar pairs.

ρ	Gmag	$\Delta Gmag$	$\Delta \xi$	$\Delta \eta$	$\Delta \xi_1$	$\Delta \eta_1$	$\Delta \xi_2$	$\Delta \eta_2$	σ_ξ	σ_η
arcsec	Mag		arcsec							
5.8	14.4	1.0	−0.045	−0.017	0.021	0.015	0.066	0.032	0.013	0.013
6.3	15.6	1.5	−0.014	−0.008	0.007	−0.013	0.021	−0.004	0.018	0.018
9.8	15.6	0.3	0.009	0.035	−0.078	0.026	−0.088	−0.008	0.019	0.019
11.3	13.3	1.4	0.009	−0.017	−0.103	−0.022	−0.111	−0.005	0.018	0.018
13.6	14.9	0.9	−0.020	−0.014	0.119	0.016	0.139	0.031	0.023	0.023

Description of the columns is given in the text (see Subsection 4.3).

Table 2

Ephemerides of the apparent close approaches observed.

NEA	T_c	$\Xi(T_c)$	$H(T_c)$	$\dot{\Xi}(T_c)$	$\dot{H}(T_c)$	mag	$D(T_c)$	Gaia source id	Gmag
		Arcsec		arcsec/hr			A.U.		
243025 (2006 UM216)	2019-11-24T00:51:29.1	11.552	−1.793	−13.9	−88.6	16.3	0.466	3365960063080786816	16.150
243025 (2006 UM216)	2019-11-24T01:38:35.6	−10.130	1.572	−13.9	−88.7	16.3	0.466	3365959960001541120	14.563
162723 (2000 VM2)	2019-11-24T19:24:44.9	1.765	−10.209	−472.8	−82.2	15.7	0.164	1942346874552792320	15.422
333555 (2005 VY17)	2019-12-10T23:40:27.2	10.033	1.811	−31.9	178.8	16.4	0.361	3436496968115120512	16.223
333555 (2005 VY17)	2019-12-11T00:53:44.0	−14.053	−2.524	−31.7	178.6	16.4	0.361	3436506073445421952	15.441
35107 (1991 VH)	2020-02-05T01:47:56.1	−5.716	−2.438	−52.0	122.1	15.6	0.310	3870223892606361856	15.583
65690 (1991 DG)	2020-03-23T19:27:54.4	4.982	2.070	−130.4	315.6	16.0	0.108	607072019309754624	13.996
65690 (1991 DG)	2020-03-23T19:36:07.5	5.646	2.345	−130.4	315.6	16.0	0.108	607072088029231232	15.972

 $D(T_c)$ is the topocentric distance to the NEA at the central moment of the approach. Description of other columns is given in the text (see the beginning of Section 5).**Table 3**

The results of observations of the apparent close approaches performed in the Pulkovo program framework.

NEA	t_c	$t_c - T_c$	$\xi(t_c)$	$(O-C)_\xi$	$\eta(t_c)$	$(O-C)_\eta$
		sec	arcsec			
243025 (2006 UM216)	2019-11-24T00:51:25.9 ± 0.6	−3.216	11.551 ± 0.011	−0.000	−1.793 ± 0.012	−0.000
243025 (2006 UM216)	2019-11-24T01:38:34.7 ± 0.9	−0.930	−10.149 ± 0.012	−0.020	1.577 ± 0.019	0.004
162723 (2000 VM2)	2019-11-24T19:24:46.4 ± 0.6	1.546	1.754 ± 0.050	−0.007	−10.201 ± 0.055	0.006
333555 (2005 VY17)	2019-12-10T23:40:19.0 ± 0.9	−8.083	10.047 ± 0.031	0.014	1.807 ± 0.033	−0.003
333555 (2005 VY17)	2019-12-11T00:53:36.5 ± 1.2	−7.744	−13.935 ± 0.039	0.097	−2.511 ± 0.041	0.014
35107 (1991 VH)	2020-02-05T01:47:54.1 ± 1.3	−1.945	−5.771 ± 0.034	−0.054	−2.471 ± 0.030	−0.033
65690 (1991 DG)	2020-03-23T19:27:54.5 ± 0.8	0.088	4.899 ± 0.057	−0.084	2.028 ± 0.052	−0.043
65690 (1991 DG)	2020-03-23T19:36:07.7 ± 0.4	0.150	5.542 ± 0.022	−0.105	2.292 ± 0.028	−0.053

Description of the columns is given in the text (see Section 5).

Table 4Apparent velocities (μ_ξ , μ_η) of the NEAs derived from the Pulkovo observations of appulses.

NEA	μ_ξ	μ_η
	arcsec/hr	
243025 (2006 UM216)	−13.7437 ± 0.0003	−88.5307 ± 0.0001
162723 (2000 VM2)	−471.8398 ± 0.0624	−81.1388 ± 0.0571
333555 (2005 VY17)	−31.9105 ± 0.0027	178.4210 ± 0.0035
35107 (1991 VH)	−52.2931 ± 0.0002	122.0332 ± 0.0002
65690 (1991 DG)	−130.6771 ± 0.0003	315.5058 ± 0.0005

The differences $t_c - T_c$, $(O-C)_\xi = \xi(t_c) - \Xi(T_c)$, $(O-C)_\eta = \eta(t_c) - H(T_c)$ may help to analyze systematic trends of the residuals in observations and/or ephemerides.

Table 4 contains the values of the components of the NEA apparent velocities calculated according to expressions from Subsection 4.4.

As can be seen from Table 3, the standard errors of the relative positions are in the 10 mas – 60 mas range. The mean value of the standard error is 30 mas for both coordinates over all events. The $(O-C)$ values are usually smaller than 60 mas in RA and 50 mas in Dec. Comparison of the $(O-C)$ values with the corresponding standard errors leads to the conclusion that the residuals lie in $1-\sigma$ – $2-\sigma$ interval in RA and $1-\sigma$ in Dec for 5 events. The exceptions are both close approaches of 1991 DG and one event for 2005 VY17. In these cases the absolute values of $(O-C)$ in

RA are close to 100 mas and exceed a $2-\sigma$ level. The central moment $(O-C)$ values are negative for almost all events and reach ≈ -8 s.

The $(O-C)$ value analysis reveals that the agreement between ephemerides and observations depends on the asteroid. The $(O-C)$ values in RA are slightly bigger than those in Dec. A possible explanation is that the ‘star – NEA’ separation at central moments of the events is systematically larger in RA than in Dec (see Table 2). Two close approaches of 1991 DG and one event for 2005 VY17 are the cases for which the systematic differences are significant. For 1991 DG we can link observed positional systematic trends with the smallest topocentric distance among all studied events. Hence, this may be interpreted as a result of the imperfection of the ephemerides. Usually the maximum approach between a star and a NEA happens before the ephemerides are predicted. The $t_c - T_c$ difference is usually about 1–2 s (that is close to the time determination uncertainty) and reach up to 8 s for both 2005 VY17 events. This effect may be caused by both unaccounted systematic trends and/or ephemerides imperfection.

6. Conclusions

The realization of the astrometric observational program of the apparent close approaches between near-Earth asteroids and the Gaia stars has started at the Pulkovo Observatory. The ‘Saturn-1m’ was used to perform the CCD-observations of eight events. The necessary technique

Table 5

The (O–C) residuals of five NEAs resulted from 2019 to 2020 observations extracted from the MPC database.

NEA	(O–C) _{α} arcsec	σ_α	(O–C) _{δ}	σ_δ
243025 (2006 UM216)	–0.073	0.263	0.088	0.117
162723 (2000 VM2)	0.092	0.559	0.087	0.312
333555 (2005 VY17)	0.021	0.084	–0.015	0.067
35107 (1991 VH)	0.006	0.217	0.098	0.426
65690 (1991 DG)	0.101	0.443	–0.027	0.490

of observations and data reduction was developed. As a result, the parameters of the approaches were obtained. The ‘star – NEA’ relative positions at the central moments of the events were calculated with the mean accuracy of 30 mas.

Table 5 allows us to provide another verification of our technique. This table contains positional (O–C)s and their standard errors for the same five NEAs whose appulses were observed according to this study. A set of NEA observations from Table 5 was extracted from the Minor Planet Center (MPC) database (https://minorplanetcenter.net/db_{\mathit{search}}) in the week-size intervals relative to the Pulkovo observations analyzed in this paper.

The (O–C)s are the mean positional residuals, σ_α and σ_δ are the standard errors of (O–C). ‘C’-positions were calculated using JPL’s HORIZONS system (<https://ssd.jpl.nasa.gov/horizons.cgi>) via astroquery-based script.

Comparison of the (O–C) values and the accuracy estimates obtained in this study (Table 3) with the results of the analysis of observations of the same NEAs presented in the MPC database (Table 5) shows the advantage of the observations of the apparent close approaches between the NEAs and the Gaia stars. The results of the performed observations show that it is possible to significantly reduce systematic errors and increase the accuracy.

As a result of the performed work, we can conclude that the astrometric observations of the apparent close approaches to the Gaia stars allows us to reach **high positional** accuracy (up to **30 mas**). The regular observations of hundreds of those events will help to collect a quality material which can be used for dynamical studies of the NEAs.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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